

HW 1

Problem 1

c)

For $n > 3$

$$D_n = \{e, x, x^2, \dots, x^{n-1}, y, yx, yx^2, \dots, yx^{n-1}\}$$

$$x = \frac{360^\circ}{n}$$

Consider:

$$R = \begin{pmatrix} \cos\left(\frac{360^\circ}{n}\right) & -\sin\left(\frac{360^\circ}{n}\right) \\ \sin\left(\frac{360^\circ}{n}\right) & \cos\left(\frac{360^\circ}{n}\right) \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$SR^j = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ \sin\left(\frac{360^\circ j}{n}\right) & \cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

Let $0 \leq j < n$

$$R^{-j}S = \begin{pmatrix} \cos\left(-\frac{360^\circ j}{n}\right) & -\sin\left(-\frac{360^\circ j}{n}\right) \\ \sin\left(-\frac{360^\circ j}{n}\right) & \cos\left(-\frac{360^\circ j}{n}\right) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Since $\cos(-\theta) = \cos(\theta)$ and $\sin(-\theta) = -\sin(\theta)$, this becomes:

$$R^{-j}S = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & \sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

The matrix multiplication gives:

$$= \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ -\sin\left(\frac{360^\circ j}{n}\right) & -\cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix}$$

which is the same as the left-hand side.

Thus, we have shown that:

$$SR^j = R^{-j}S$$

To show $R^{-j} = R^{n-j}$, consider: Let $0 \leq j < n$

$$R^{-j} = \begin{pmatrix} \cos\left(-\frac{360^\circ j}{n}\right) & -\sin\left(-\frac{360^\circ j}{n}\right) \\ \sin\left(-\frac{360^\circ j}{n}\right) & \cos\left(-\frac{360^\circ j}{n}\right) \end{pmatrix}$$

$$R^{n-j} = \begin{pmatrix} \cos\left(\frac{360^\circ(n-j)}{n}\right) & -\sin\left(\frac{360^\circ(n-j)}{n}\right) \\ \sin\left(\frac{360^\circ(n-j)}{n}\right) & \cos\left(\frac{360^\circ(n-j)}{n}\right) \end{pmatrix}$$

Noting that:

$$\cos\left(\frac{360^\circ(n-j)}{n}\right) = \cos\left(\frac{360^\circ n}{n} - \frac{360^\circ j}{n}\right) = \cos\left(-\frac{360^\circ j}{n}\right)$$

$$\sin\left(\frac{360^\circ(n-j)}{n}\right) = \sin\left(\frac{360^\circ n}{n} - \frac{360^\circ j}{n}\right) = -\sin\left(\frac{360^\circ j}{n}\right)$$

we have:

$$R^{n-j} = R^{-j}$$

Subgroup of $GL_2(\mathbb{R})$: $\langle R, S \mid R^n = S^2 = I, SR^j = R^{n-j}S \rangle = A$

- (Associativity) Given by matrix multiplication
- (Identity)

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

this is a zero-degree rotation

- (Inverses) Each rotation matrix R^k s.t. $1 \leq k < n$ has an inverse R^{n-k} . Each reflection matrix SR^j s.t. $0 \leq j < n$ has an inverse itself.

$$R^k R^{n-k} = R^n = I = R^n = R^{n-k} R^k$$

$$SR^j SR^j = SR^j R^{n-j} S = SIS = SS = I$$

- (Closure)

1. Let R^j and R^k s.t. $1 \leq j, k < n$

$$R^j R^k = \begin{pmatrix} \cos\left(\frac{360^\circ j}{n}\right) & -\sin\left(\frac{360^\circ j}{n}\right) \\ \sin\left(\frac{360^\circ j}{n}\right) & \cos\left(\frac{360^\circ j}{n}\right) \end{pmatrix} \begin{pmatrix} \cos\left(\frac{360^\circ k}{n}\right) & -\sin\left(\frac{360^\circ k}{n}\right) \\ \sin\left(\frac{360^\circ k}{n}\right) & \cos\left(\frac{360^\circ k}{n}\right) \end{pmatrix}$$

Using the angle addition formulas $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$ and $\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$

$$= \begin{pmatrix} \cos\left(\frac{360^\circ j+k}{n}\right) & -\sin\left(\frac{360^\circ j+k}{n}\right) \\ \sin\left(\frac{360^\circ j+k}{n}\right) & \cos\left(\frac{360^\circ j+k}{n}\right) \end{pmatrix}$$

This is another rotation matrix specifically $R^{j+k \mod n}$

2. Let SR^j and R^k be a reflection and a rotation s.t. $0 \leq j < n$ and $1 \leq k < n$. Then

$$(SR^j)R^k = SR^{j+k \mod n}$$

This is another reflection in the group.

$$R^k(SR^j) = (R^kS)R^j = S(R^{n-k}R^j) = SR^{n+j-k \mod n}$$

This is another reflection in the group.

3. Let SR^j and SR^k be reflections s.t. $0 \leq j, k < n$

$$SR^jSR^k = R^{n-j}SSR^k = R^{n+k-j \mod n}$$

This is another rotation in the group.

Let $\varphi : D_n \mapsto A$ s.t. $\varphi(x^j) = R^j$ s.t. $1 \leq j \leq n$, $\varphi(y) = \varphi(S)$, and $\varphi(yx^j) = SR^j$ s.t. $1 \leq j < n$

Well Defined: Le

Homomorphism: Let $a, b \in D_n$

1. If $a = x^j$ and $b = x^k$ s.t. $1 \leq j, k \leq n$
 $\varphi(x^jx^k) = \varphi(x^{j+k}) = R^{j+k} = R^jR^k = \varphi(x^j)\varphi(x^k)$
2. If $a = x^j$ and $b = yx^k$ s.t. $0 \leq j, k < n$
 $\varphi(x^jyx^k) = \varphi(yx^{n-j}x^k) = SR^{n+k-j} = R^jSR^k = \varphi(x^j)\varphi(yx^k)$